

Tech. Comm. Skills (SS2011) Final Exam

Date: May 21, 2026

Course Instructors:

Dr. Saadia Rasheed

Ms. Zakia Naeem

Ms. Areeba Nadeem

Total Time (Hrs.): 3

Total Marks: 75

Total Questions: 6

Roll No

EE 4 A,B,C
Sections

Do not write below this line.

Attempt all the questions. Do not solve on the question paper. DO NOT change the actual Question Numbers. You may spend about 50 – 55 minutes on each part. Use Bloom's Taxonomy vocabulary.

PART A: CLO 3: Demonstrate critical reading skills to evaluate the use of statistical tools and conclude findings effectively.

Q1. Scenario: Conversion of written information into an infographic.

Instructions: Read the following information and transform it into a clear and well-labeled infographic. You may choose a suitable format such as a pie chart, bar graph, or any other diagram.

Power Loss in Transmission System

In an electrical transmission system, total power loss is distributed among different factors. Line resistance accounts for 35% of losses, transformer inefficiency contributes 25%, corona loss stands for 15%, leakage and insulation faults account for 15%, and other technical losses make up the remaining 10%. Reducing resistance and improving transformer efficiency can significantly enhance overall system performance. [10]

Q2. Scenario: Data interpretation through a table. [15]

Analyze the given table. Compare energy sources based on power generation and cost efficiency.

✓ Categorize them into renewable and non-renewable groups, and conclude which sources are most efficient for future energy planning. Compose your findings in a clear, cohesive, and objective written form. Use proper technical terminology. (100 – 120 words).

"Power Generation and Cost Analysis"

Energy Source	Power Generated (MW)	Type	Cost per Unit (PKR)
Thermal	800	Non-Renewable	18
Hydropower	500	Renewable	06
Solar	300	Renewable	10 -
Wind	200	Renewable	09
Nuclear	200	Non-Renewable	14

Power Gen }
Cost eff }
↓
Renew Non
Most eff?

PART B: CLO 4: Identify workplace conflict and provide resolution strategies.

Q3. Read the given case study and answer the questions that follow.

[3*5=15]

The VoltNet Project Team

VoltNet Solutions assigned a five-member team to complete a client project within eight weeks. The team included project manager Nadia, engineers Omar, Priya, and Tariq, and quality specialist Leila. Their task was to build a working product, prepare a written report, and deliver a live presentation to the client.

Before the first meeting, Nadia emailed an agenda, a budget summary, and a set of questions for members to consider. When the meeting began, Omar admitted he had not read the email. Tariq had only glanced at it. Leila had not looked at the attachments at all. Only Jin, a software team member added later, arrived fully prepared with notes. Unable to hold a useful discussion, Nadia spent the whole session just explaining the project scope, time that should have been used to plan and divide tasks.

As the project moved forward, each member naturally settled into a role. Jin regularly shared research findings and raised concerns about gaps in the client's requirements. When Omar and Jin argued over the best approach to take, Tariq calmly helped both sides understand each other and find a middle ground. Leila kept detailed notes after every meeting and shared them with the group the next day. Priya, though rarely the loudest voice in meetings, quietly managed schedules, booked rooms, and made sure the team had everything they needed before each work session.

In Week 4, the team agreed to first define what a good solution must look like before suggesting any ideas. They set clear requirements: the solution had to meet the client's performance standard, work within the existing system, and stay within budget. Only after agreeing on these standards did they brainstorm. They generated over a dozen ideas and then filtered them against their criteria. Through open discussion, all members agreed on a final approach that everyone felt comfortable supporting.

In the final week, Leila and Priya prepared the written report while Jin created a visual slide summary for the client. Omar and Tariq led the presentation, with Nadia guiding the session as moderator. The client was satisfied and accepted the work with only small changes requested. The team later agreed that their success came not from one strong leader, but from every member playing their part at the right time.

- Analyze Tariq's contributions to the team's progress and examine how his specific role directly influenced the group's success.
- Compare the potential outcomes of public versus private confrontation and justify which approach would best preserve team morale and productivity.
- Illustrate a complete checklist of participant responsibilities by applying the 'before, during, and after' framework to a professional engineering meeting scenario.
- Evaluate the performance of both the leader and the participants; defend your position on who bears primary responsibility by providing specific professional standards for each.
- Assess whether consensus method was the right choice, or contrastingly, a majority vote method would have worked better for the VoltNet team's goals.

NOTE: Do not overexplain or become too generic. Be precise and clear.

Q4. Explain why conflict is important in groups. Describe any one type of conflict in groups and provide two communication strategies to manage them effectively.

[3+2+5=10]

Spring 2026

Department of Electrical Engineering

Page 2 of 3

National University of Computer and Emerging Sciences
Lahore Campus

PART C: CLO 5: Demonstrate job seeking and interview skills to secure employment opportunities.

Q5. Generate specific and clearly outlined answers to all the following questions: Each question carries three marks. Avoid generic answers. Use specific vocabulary and write clear sentences. [15]

- Illustrate five key Do's and five Don'ts one should consider and follow during a job interview?
- During interview responses, how can you **explain** your group lead's technical role-impact on any of the university projects that you were a part of?
- As an electrical engineering student, **differentiate** between your technical skill (*specify only one*) and your developing professional area (*specify only one*). **Analyze** how each impacts your suitability for this specific job role.
- Which mode of interviewing would you prefer to choose: online or face-to-face? **Justify** your answer with two strong reasons.
- How can you **apply** the technical lessons learned in your final year project to solve a real-world problem?

Q6. In an interview situation, the interviewer asks you to **elaborate** any three techniques you will use to convince a client to shift from the conventional to renewable energy sources. (**incorporate ethos, logos and pathos** to develop your arguments for each to persuade the client). [10]

